



OWL
Cyber Defense

Owl Cyber Defense Government Solutions

Raise the Bar (RTB) and High Threat Network (HTN) Requirements



OWL Cyber
Defense

+

TRESYS

- Hardware expertise – data diode market leader
- Broad commercial and government/DoD market experience
- Comprehensive product lines designed to meet a variety of use cases
- ISO 9001 certified, U.S.-based supply chain and manufacturing
- Software/OS hardening expertise - SELinux policy implementation
- Government tested, vetted and accredited Cross Domain Solutions with Deep Content Inspection & leading filtering
- Deep ties to the intel and DoD communities - a true government partner
- Extensive professional services portfolio for accreditation customization and training



New Guidelines for CDSs and One-Way Transfer Systems (aka Diode Solutions)

- 2018 Raise the Bar (RTB)
Baseline Release Version 1.2 4
April 2019
- 2019 Raise the Bar (RTB)
Baseline Release Version 2.1 7
April 2020
- 2020 Raise the Bar (RTB)
Baseline Release Version 3.0 2
November 2020





Hardware-enforcement for cross domain transfers with HTN touchpoints isn't a want or need – **IT IS THE MANDATE**

Important Dates to Remember:

- **BY DECEMBER 2021:** All unclassified CDS network connections will require hardware enforced one way transfer mechanisms, diodes
- **BY DECEMBER 2022:** All SECRET Releasable CDS network connections will require hardware enforced one way transfer mechanisms, diodes

Implications:

- Mission risk due to evolving nation state cybersecurity threats
- Authority To Operate (ATO) could be compromised
- Operational, schedule and connection impact



- Tested to the most rigorous US government standards – LBSA
- NCDSMO Baseline CDS
- MDA FMS CDS Listing
- NSA Diode Listing
- Only US Government approved media transfer device



- Accreditation and Certification
- Security Architecture Assessment
- Product Security Analysis
- Platform Security
- Custom Data Filtering
- On-site Installation Services
- SE Linux Training



Bi-Directional CDSs



XD BRIDGE™

- LBSA Certified
- ITAR Controlled CDS
- NCDSMO Baseline
- Common Criteria EAL4+ OS w/SELinux
- Operationally Accredited
- 1U Data Center Appliance
- Uni- and Bi-Directional
- TCP FTP/FTPS/SFTP File Transfers
- UDP Streaming Messages



XD GUARDIAN™

- LBSA Certified
- Exportable CDS
- FMS Baseline
- Common Criteria EAL4+ OS w/SELinux
- Operationally Accredited
- 1U Data Center Appliance
- Uni- and Bi-Directional
- TCP FTP/FTPS/SFTP and HTTP/REST File Transfers
- UDP Streaming Messages
- DFDL/XML Filtering

Uni-Directional Simple Diode Solutions - XD PRISM



- Operationally Accredited
- Uni-Directional
- Data Center 1U/2U Servers and 1U Appliance

- Small Form Factor and Ruggedized Tactical Appliances
- RFTS/DFTS File Transfers
- UDP/TCP Streaming Messages



- Air Force Weather Agency: move to the Cloud in 2025
- Global Strike Command Support
- Joint Strike Fighter (JSF)
- Rivet Joint Program
- Minuteman III: Telemetry Data
- Support Missions out of AFRL



Diodes/Diode-Based Platforms

Common Criteria

V7 (EAL 4+ - Active)
010 (EAL 2 - Active)
Fusion V5 (TBD)

NSA Diode Listing

Scowt-Compact (010 -TBD)
Scowt (V7 MC- Active)
OCDS-1000 (V7t - Active)
Fusion V3 (ICEPIC V3 – In Progress)
Fusion V5 (ICEPIC V5 - TBD)
OCCS Kit (V7 HC - Active)

OS Certifications

Common Criteria, FIPS 140-2, NIAP

RedHat RHEL 6.x (Active)
RedHat RHEL 7.6 (In Progress)

Cross Domain Solutions (NCDSMO)

NCDSMO/LBSA Baseline

XD Bridge V3 FT-TSC (JSF) (Active)
XD Bridge ST/FMV/DCI (In LBSA)
XD Bridge V3 IP-TSC (Sunset)
XD Bridge V3 FT-TSC (Sunset)
OCDS ST/FT (Sunset)
ECDS ST/FT (Sunset)

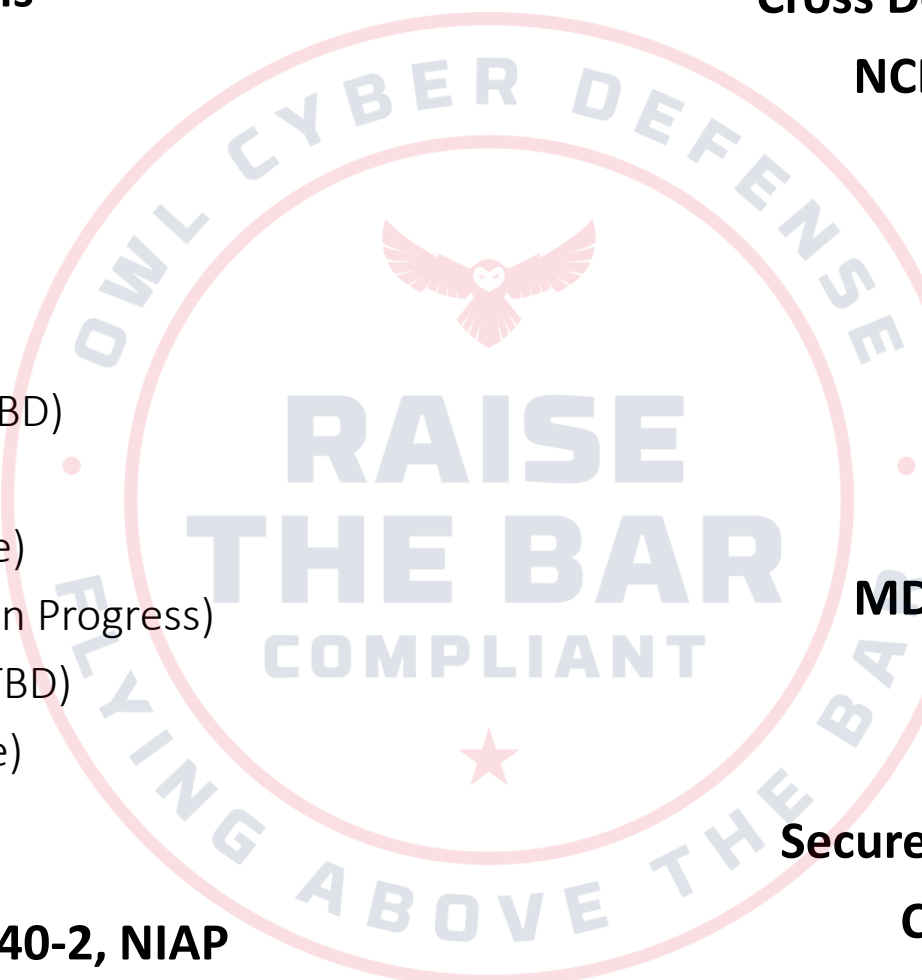
MDA/FMS/LBSA Listing

XD Guardian FT-TSC (Active)
XD Guardian XML-TSC (In LBSA)

Secure Media Content Transfer

CyberCommand, Army, Air Force

XD Air





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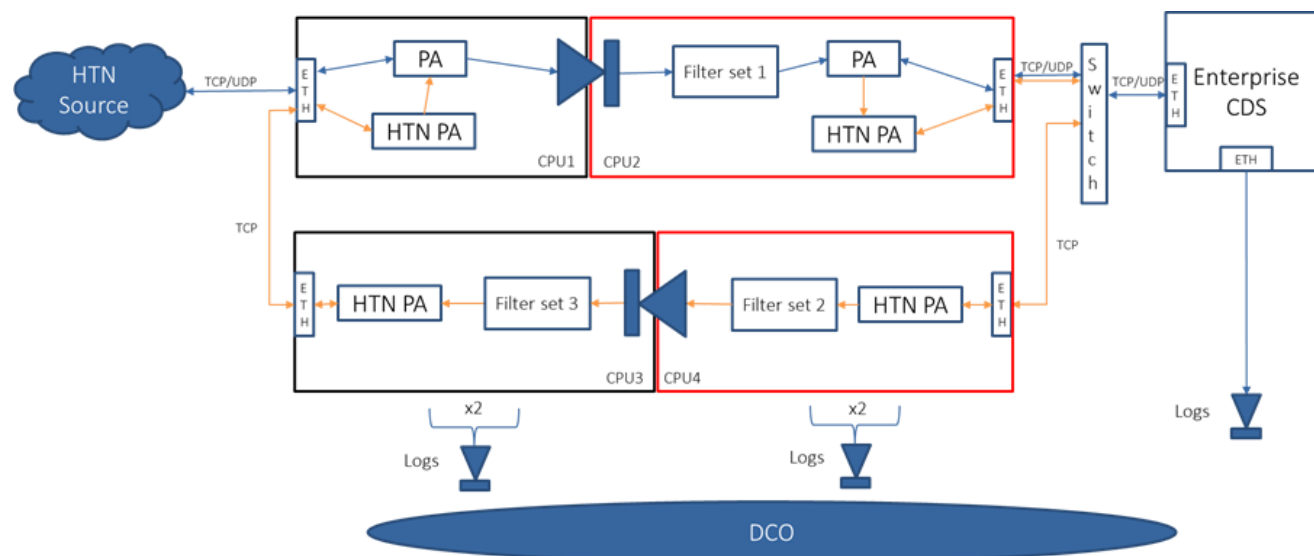
Thank You!



Additional Information



Two XD Bridge ST-TSC solutions to support bi-directional dataflows between High Threat Network (HTN) and multi-domain CDS



2021 GA

TCP-based HTN Protocol Adapter for XD Bridge-to-XD Bridge dataflow



A High Threat Network is defined as a network in which a known or suspected highly skilled (e.g. nation-state or transnational organized crime level) cyber actor is operating.

The cyber actor may be operating on the network without the knowledge of the primary user of the network.

HTNs are also networks for which the cyber hygiene is known to be insufficient, the entities on the network are known not to have robust and effective Defensive Cyber Operations capabilities, and the lack of ownership of the network or ability to influence the security posture of it.

An organization's Authorizing Official in consultation with USCYBERCOM and the USG Intelligence Community will determine if a classified network is to be considered high threat.

Unclassified networks connected to the Internet (directly) or indirectly (NIPRNet) are considered high threat.



New Guidelines for CDSs and One-Way Transfer Systems (aka Diode Solutions)

- 2018 Raise the Bar (RTB) Baseline Release Version 1.2, 4 April 2019
- 2019 Raise the Bar (RTB) Baseline Release Version 2.1, 7 April 2020
- 2020 Raise the Bar (RTB) Baseline Release Version 3.0, 2 November 2020
- One-Way Transfer Design Patterns
- Trusted Software Base with Hardware Assist (THA) Design Patterns – Traditional software-enforced CDS reinforced with hardware-enforced diodes.
- Establishes need for robust filtering solution that is required for content inspection and sanitization
- Expands NCDSMO Lab Base Security Assessment (LBSA) evaluations for CDSs
- Establishes new NSA Diode evaluation and approval process (separate from CDS LBSA)
- Driving changes to Commercial Solutions for Classified (CSfC) Capability Packages: Continuous Monitoring, Mobile Access, Multi-Site Connectivity with new one-way data transfer guidelines



Any High Threat Network (HTN) (All NIPR/Internet and some Classified) connections requires the use of Hardware-Enforced one-way transfer mechanism

RTB Phase 1: End of CY2020 (GAR-2, OWTDP-4a, OWTDP-4b)

RTB Phase 2: End of CY2021 (CUCR, OWTDP-4c)

RTB Increment B: End of CY2021 (ECDSP-Uni/Bi-Directional and P2P/Tactical-Unidirectional)

RTB Increment C: End of CY2022 (P2P/Tactical-Bidirectional)

RTB Increment D: End of CY2023 Hardware-based filtering (ECDSP)

RTB Increment E: End of CY2024 Hardware-based filtering (P2P/Tactical)

- Intent is to require RTB-compliant filtering that satisfies Rule of Three on any traffic flow that establishes round trip to HTN
- Impacts Enterprise Cross Domain Service Providers (ECDSP), Point-to-Point (P2P), and Tactical deployments
- Includes one-way and two-way (aka bi-directional) One-Way Transfer Design Patterns (OWTDP) identifying combining Diode Solutions, THA CDSs, and Multi-Domain CDSs for transfers involving all security domains connecting with HTN.



Certifiable Linux Integration Platform (CLIP)

- No longer Government funded project
- Internal R&D
- On-going updates
- SELinux RefPolicy consistency
- Active efforts: SELint (2020)
 - Designed to perform static code analysis on Reference Policy (refpolicy) SELinux source code. SELint is designed to help developers write policy that is more maintainable, readable, and secure, and reduce the time spent debugging challenging policy issues.

Data Format Description Language (DFDL)

- Active Government funded project
- Number of DFDL schemas rapidly expanding
- Addressing backlog of DoD and IC structured data messaging constructs
- Technology maturation

Hardware Enforcement

- Internal R&D
- FPGA Diode with performance and form factor improvements
- FPGA filtering
- Embedded OEM components



ST-TSC
For transferring, with limited latency, messages and files that contain structured and fixed format data using linear filtering pipeline



DCI-TSC
For transferring files that contain complex, structured, and fixed format data using filter orchestration and recursive decomposition filtering pipeline.



FMV-TSC
For transferring, with limited latency, full motion video using linear filtering pipeline

IP-TSC
For transferring UDP traffic through IP gateway.

FT-TSC
For transferring files using limited filtering pipeline.





Owl provides customers the most experienced hardware-enforcement and overall cross domain security technology expertise

- Uni-directional, Bi-directional
- Hardware- and Software-Enforced
- Diode Solution CDSs and Simple Diode Solutions
- Fixed Format, Structured and Complex Data, and Video filtering pipelines
- Broad range of Data Center and Tactical platforms